

ALEXIS BASE

Florida Atlantic University 5600 US-1, Fort Pierce, FL
www.linkedin.com/in/alexis-base/ • abase2015@fau.edu • (561) 221-4323

EDUCATION

Florida Atlantic University- Harbor Branch Oceanographic Institute, Fort Pierce, FL

Doctor of Philosophy in Ocean Engineering, Fall 2023-Present

Ph.D. Candidate

Dissertation Title: In situ imaging systems for characterization of particles, plankton and biophysical interactions in diverse aquatic environments

Florida Atlantic University, Boca Raton, FL

B.S. Ocean Engineering, Spring 2023

Minor in Mathematics

M.S. Ocean Engineering, Summer 2024

FAU High School, Boca Raton, FL (2019)

Relevant Coursework: Ocean Optics, Computational Foundations of AI, Advanced Hydrodynamics, Acoustics, Physical Oceanography, Signal Processing, Engineering Data Analysis, Biological Oceanography

JOURNAL PUBLICATIONS

Dass, S., **Base, A.**, Agarwal, K. and Nayak, A.R. (2026), An in situ stereo-imaging system for characterizing behavior, distributions, and aggregations of meso- and macro-zooplankton in diverse aquatic environments. *Limnol Oceanogr Methods* e70043. <https://doi.org/10.1002/lom3.70043>

D. Sanborn, **A. Base**, L. Garavelli, R. Barua, J. Hong, and A.R. Nayak. 2023. Digital holography for real-time non-invasive monitoring of larval fish at power plant intakes. *Canadian Journal of Fisheries and Aquatic Sciences*. **80**(9): 1470-1481. <https://doi.org/10.1139/cjfas-2023-0058>

A. Base, K. Agarwal, A.R. Nayak, "AUTOHOLO: a modular in situ digital holographic imaging system for diverse aquatic applications" **(In Preparation)**

RESEARCH EXPERIENCE

Graduate Research Assistant, Florida Atlantic University- Harbor Branch Oceanographic Institute

Fort Pierce, FL

Advisor: Dr. Aditya R. Nayak

2022 – Present

- Conducting laboratory and field research on a variety of research projects with a focus on in-situ instrumentation for aquatic biophysical studies.
- *Key Instruments Utilized:* (holographic microscopy) AUTOHOLO, custom built stereo-vision system, and in-situ acoustic instruments

Independent Research Project, Florida Atlantic University

Mentor: Dr. Myeongsub (Mike) Kim; Microfluidics and Energy Laboratory

October 2017- August 2020

"A Synthetic Sea Urchin Exoskeleton for Carbon Conversion to Environmental Calcium Carbonate Precipitates"

- Worked on the development of a novel solution to reduce carbon dioxide pollution and ocean acidification by biomimicking Sea Urchins' ability to convert CO₂ into Calcium Carbonate by lowering the concentration of CO₂ in flue gas from powerplants

Independent Research Project, Florida Atlantic University

October 2014 - March 2017, August 2019 - August 2020

“C.O.D.E. Computer Operated Detection Environment for Identifying the Invasive Red Lionfish in South Florida and the Bahamas”

- Programed and developed an image recognition algorithm to identify Red Lionfish underwater to study their population size and limitations
- Worked on a motor control sequence to correspond to the image program using C++ and Python on the Raspberry Pi
- Utilized a modified The Blue Robotics BlueROV for implementation as an AUV.

RESEARCH FIELD EFFORTS (Selected)

- R/V Hogarth, Gulf Stream Research Cruises (Mar 20, 2025; Apr 21, 2023)
- R/V Ira C and R/V Nucella, Damariscotta River Estuary (Summers 2022-2024)
- R/V Aranda, Baltic Sea Research Cruise (Apr 16-26, 2024)
- R/V SOCIB, Balearic Sea Research Cruise (Mar 7-9, 2023)

CONFERENCE/SYMPOSIUM PRESENTATIONS (GRADUATE)

O. Ruchti, **A. Base**, K. Spilling, A. R. Nayak “Exploring Community Dynamics and Environmental Drivers in the Baltic Sea Using High-Resolution In Situ Holographic Imaging”, Feb 22-27 2026, in Glasgow, Scotland.

A. Base, S. Dass, K. Agarwal, A.R. Nayak, “An In Situ Stereo Imaging System for Zooplankton Behavioral Studies in the Indian River Lagoon”, Feb 19, 2026. Indian River Lagoon Symposium, Fort Pierce, Florida, USA.

A. Abdullah, O. Ruchti, **A. Base**, A. R. Nayak “An In Situ Digital Holographic Imaging System (AUTOHOLO) to Characterize Cyanobacterial Blooms in Freshwater and Estuarine Environments ”, Feb 19, 2026. Indian River Lagoon Symposium, Fort Pierce, Florida, USA.

M. E. Bennett, **A. Base**, A. R. Nayak, “Advancing Plankton Monitoring in the Indian River Lagoon: A Comparative Study of the Imaging FlowCytobot and AUTOHOLO” Feb 19, 2026. Indian River Lagoon Symposium, Fort Pierce, Florida, USA.

S. Dass, **A. Base**, K. Agarwal, A.R. Nayak, “Development of a custom-built stereo-imaging system for in situ zooplankton studies”, Mar 26-31, 2025. ASLO Aquatic Sciences Meeting, Charlotte, North Carolina, USA.

A. Base, S. Dass, K. Agarwal, A.R. Nayak, “An In Situ Stereo Imaging System for Zooplankton Behavioral Studies in the Indian River Lagoon”, Feb 20-21, 2025. Indian River Lagoon Symposium, Fort Pierce, Florida, USA.

A. Base, S. Dass, A.R. Nayak, “Characterizing in situ small scale biophysical interactions of the mysid shrimp, *Neomysis americana*, using stereoimaging”, Feb 18-23, 2024. Ocean Sciences Meeting, New Orleans, Louisiana, USA.

S. Dass, **A. Base**, A.R. Nayak, “An in situ stereo imaging system for characterizing zooplankton distribution and behavior”. Feb 18-23, 2024. Ocean Sciences Meeting, New Orleans, Louisiana, USA.

O. Ruchti, **A. Base**, A.R. Nayak, “In situ digital holography for plankton monitoring in coastal and estuarine environments”, Nov 12-16, 2023. 27th Biennial CERF Conference, Portland, Oregon, USA.

A. Abdullah, A.U. Kerkar, R. Barua, **A. Base**, T. Moore, K. Bosse, M. Sayers, M. McFarland, A.R. Nayak, “Characterizing cyanobacterial blooms and plankton community composition in western Lake Erie using

in situ digital holography”, Oct 23-28, 2022. 11th U.S. Symposium on Harmful Algae, Albany, New York, USA.

A. Base, L. Nyman, L. Garavelli, R. Barua, A.R. Nayak, “A digital holographic imaging system for in situ monitoring of endangered and sensitive fish larvae in cooling water intake structures.”, Feb 28 – Mar 4, 2022, *Virtual Ocean Sciences Meeting*.

TEACHING AND MENTORING EXPERIENCE

Marine Advanced Technology Education ROV Competition Mentor, FAU High School, Boca Raton, FL
January 2025 – May 2025

- Mentor to 38 high school students across 3 robotics teams (FAU Night Owls, Sea Owls, Crab Company)
- Led teams to top placements in regional competitions; Sea Owls qualified for 2025 World Championship in Michigan

PSAT/SAT/ACT Instructor, Ray Dass Test Prep, Cooper City, FL
October 2018 – Present

- Teach math (Pre-Algebra to Calculus) and test prep strategies to middle and high school students
- Instruct up to 1,000 students virtually across the U.S.; lead live sessions at private schools
- Developed customized curricula in math and reading

Peer Mentor, FAU’s Office of Undergraduate Research and Inquiry, Boca Raton, FL
August 2017 – August 2021

- Created and led workshops to promote undergraduate research engagement
- Collaborated with peer mentor team to increase OURI participation through outreach events

Education Outreach Project, “Developing a Novel Solution to Lionfish Invasion: Teaching Middle School Students Ocean Engineering and Marine Biology” December 2016 - February 2017

- Designed and delivered marine biology and ocean engineering lessons to middle school students
- Assessed program effectiveness in promoting STEM interest

3D Printing Troubleshooting and Maintenance Counselor, STEM HQ, Boca Raton, FL
Summer 2016

- Taught CAD software (OnShape, SolidWorks) and 3D printing to students (elementary–high school)
- Maintained and operated various 3D printers (MakerBot, Flash Forge, FormLabs, etc.)

INTERNSHIP EXPERIENCE

Marilyn C. Link Memorial Internship, Harbor Branch Oceanographic Institute – FAU, Fort Pierce, FL
June 2021 – August 2021

- Conducted research on a submersible holographic imaging system known as AUTOHOLO
- Studied Sturgeon fish larvae at Pacific National Northwest Laboratory (PNNL)
- Completed data analysis and processing of thousands of images using MATLAB

LEADERSHIP EXPERIENCE

Team Lead, Ocean Engineering Systems Control and Design, Bio-Inspired Self-Burying AUV
Florida Atlantic University (SeaTech) Dania Beach, FL
August 2021 – April 2022

- Designed printed circuit boards (PCBs) and electronic schematics
- Secured funding through internal grants, NEEC, and NSF support
- Led and presented weekly team meetings on technical and administrative progress

WORKSHOPS

CoPartCoFlow, Complex particle transport in complex flows with environmental applications, Jun 15-20, 2025. La londe les Maures, France (*selected applicant for small summer school program*)

Fluids and Marine Ecology Workshop, Oct 9-11, 2022. Woods Hole Oceanographic Institution (WHOI), Woods Hole, MA, USA.

FELLOWSHIPS

Indian River Lagoon Graduate Research Fellowship \$10,000 (2025), \$10,000 (2024)

FAU Summer Undergraduate Research Fellowship \$4,000 (2018)

GRANTS

FAU Office of Undergraduate Research Grant \$1,200 (2022), \$600 (2018), \$1,000 (2017)

FAU WAVE Competition Start-Up Funding \$500 (2022), \$1,000 (2019), \$500 (2017)

SCHOLARSHIPS

Dr. Steven G. Schock Scholarship for Underwater Acoustics (2026)

FAU Spirit of Atlantic Scholarship (2019)

FAU Wilkes Honors College Science Merit Scholarship (2019)

Chuck Skoch Florida Sea Grant Scholarship (2019)

Florida Academic Bright Futures Scholarship (2019)

AWARDS

FAU's Art of Science, Honorable Mention, "Stereovision Illumination" (2024)

FAU Undergraduate Research Symposium, 1st place Oral Presentation (2020), 2nd place Oral Presentation (2018), 1st place Poster Presentation (2017)

FAU WAVE, 3rd Place (2020), 2nd Place (2019)

Other Awards: Florida Association for Water Quality Control 1st Place (2019), Palm Beach Pollution Prevention Coalition (2018), INTEL Excellence in Computer Science (2017), United States Submarine Veterans Underwater Technologies Award (2017), Boca Bearing Excellence in Innovation (2016), ONR Naval Science Award (2016), NOAA Taking the Pulse of the Planet Award (2016)

IN THE NEWS

FAU Broward Campuses - Student Spotlight "**Alexis Base**" <https://www.fau.edu/broward/news-students-spotlight/alexis-base/> December 1, 2021

The Coastal Star "**Outdoors: Teen takes tech approach to stalking the lion without paws**" by Chris Felker <https://thecoastalstar.com/profiles/blogs/list/tag/alexis+base> February 4, 2015

CERTIFICATIONS

FBPE Engineer Intern (2025)

AAUS Scientific Diver (2023)

DOI MOCC Scientific Boater (2023)

DAN CPR/AED/First Aid Certified (2021, 2023)

HackerRank Python (Basic) (2024)

MEMBERSHIP

Association for the Sciences of Limnology and Oceanography, *Member* (January 2022 – Present)

Society of Naval Architects and Marine Engineers, *Member* (November 2020 – Present)

Marine Technology Society, Member (February 2023 – Present)
Marine Robotics Club, *Member* (January 2016 – May 2023)
Council for Scholarship and Inquiry, *Officer* (August 2017 – May 2023)
Council of Undergraduate Research, *Member* (August 2017 – May 2023)
Human Powered Submarine Club, *Member* (October 2019 – May 2023)
Association of STEM Entrepreneurs, *Member* (October 2019 – August 2020)